

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 112 — Past Questions & Materials
by Renance

Course: CHM 112
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 100

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Paper 1 of 2 — CHM 112 2016-2017

Level 100 • First Semester • PastQuestion

Calculate the molarity of calcium chloride (CaCl₂) in 0.5 litres of water,

x State Dalton's law of partial pressure

ie % 6, Calculate the volume of a 0.06 mol/dm³ magnesium bromide solution

moles of magnesium bromide, " ~ "# ' a

7. How many moles of a gas would be present in a gas trapped within a 37.0 liter vessel

80°C at a pressure of 2.60 atm?

8. A given mass of gas occupies a volume of 50 cm³ at temperature of 27°C. At

100°C, what will the volume occupied be?

9. How many moles of nitrogen are combined with 4.20 moles of Cl atoms in NCl₃?

10. A given mass of gas at a pressure of 2000 kPa occupies a volume of 30 cm³. What is the pressure exerted when the volume changes to 890 cm³?

11. What is the pOH of a solution with [H⁺] = 1.00 × 10⁻⁷ M?

12. Calculate the [H⁺] of a solution whose pH is 3.70

13. Calculate the pH of a buffer solution which contains the weak monoprotic acid, propanoic acid (CH₃CH₂COOH), in concentration 0.1 M and sodium propanoate in concentration 0.05 M. K_a of propanoic acid is 1.26 × 10⁻⁵ M.

14. Aqua regia is a concentrated mixture of: and HNO₃

15. In the reaction H₂SO₄ + H₂O → H₃O⁺ + HSO₄⁻, the conjugate base of H₂SO₄ is

16. Given the equilibrium AgCl(s) ⇌ Ag⁺ + Cl⁻ in aqueous solution. If solid NaCl is added to the solution, what would be the effect on the solubility of AgCl?

17. Determine the solubility product of AgBr at 25 °C if its solubility is 7.08 × 10⁻⁷ M at the same temperature.

18. Calculate the pH of a 0.1 M solution of acetic acid.

28. Calculate the current,

29. State Faraday's laws of electrolysis:

What is the current required to produce 18.0g of Aluminium in 1.50 hours?

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30. Write the half-reactions for the following:

1. CuSO₄

* 31. A dilute solution of

of electrolysis

Reaction time: - t :

solute Cu^{2+} at the anode and cathode during the electrolysis of

CuSO_4 using inert electrodes

5. Calculate the

volume of copper (II) chloride was electrolyzed using graphite electrodes.

5. of

Volume of gas was evolved at s.t.p. Calculate the mass of copper deposited ($\text{Cu} = 64.0$;

moles of gas occupies 22.4 dm^3 at s.t.p) *

732. $\text{MnO}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{MnSO}_4 + \text{H}_2\text{O} + \text{O}_2$. For this reaction, the oxidation product is

3. ;

25h, Define reducing agent in terms of electron transfer

34. Identify the oxidizing agent in the reaction: $\text{Cr}^{3+} + \text{Sn}^{4+} \rightarrow \text{Cr}^{6+} + \text{Sn}^{2+}$

38. Under which conditions would a real gas behave as ideal?

35. Define a base in the Lewis sense ; ie

~ 36. What is an acid anhydride? The anhydride of sulphuric acid is

37. What is a buffer solution?

39. The pressure exerted in a gas container is as a result of

40. An element has two isotopes; one of mass 14.003 u, the other with a mass of 15.001 u. If the relative abundance of the isotopes are 99.63% and 0.37% respectively, find the atomic mass of the element.

41. How many moles of silicon are in 4.60g of Si? ; ba a

Paper 2 of 2 — CHM 112 (physical chemistry)

Level 100 • First Semester • Handout

This source paper is a scanned document; its pages are reproduced below.

COURSE: CHM 112 (PHYSICAL CHEMISTRY)**TOPIC: ACIDS, BASES AND pH SCALE****TUTOR: KISORANKING SAM.****ACIDS AND BASES**

Before now, there is just one knowledge of the definition of acids and of bases.

But in this course, there are 3 different concepts that defined acids & bases.

1. Arrhenius Concept
2. Bronsted - Lowry Concept
3. Lewis Concept.

1. Arrhenius Concept of Acids and Bases

An Acid is any substance that gives off H^+ as its only positive ion when dissolved in water.

A Base is any substance that gives off OH^- when dissolved in water.

2. Bronsted - Lowry Concept of Acids and Bases

An acid is a Proton Donor

A Base is a Proton Acceptor

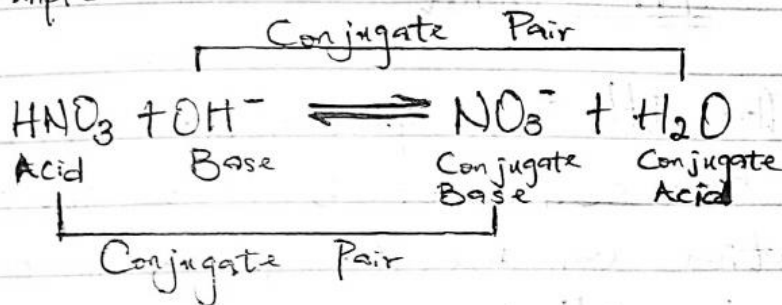
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The Bronsted Lowry Concept of acids and bases gave rise to conjugate acids and bases.

A Conjugate Acid is formed when a base accepts a proton.

A Conjugate Base is formed when an acid donates a proton.

example

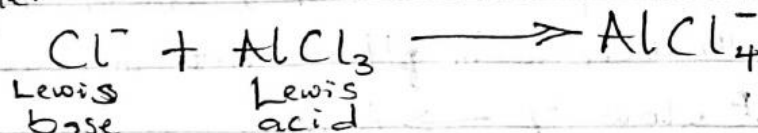


3. Lewis Concept of Acids and Bases

An Acid is that which accepts electron.

A Base is that which donates electron.

example.



NOTE: Aqua regia is a mixture of conc. HNO_3 and HCl .

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pH ANALYSIS

pH is the measure of hydrogen ion concentration in an aqueous solution. It is sometimes defined as the measure of hydroxonium ion concentration.

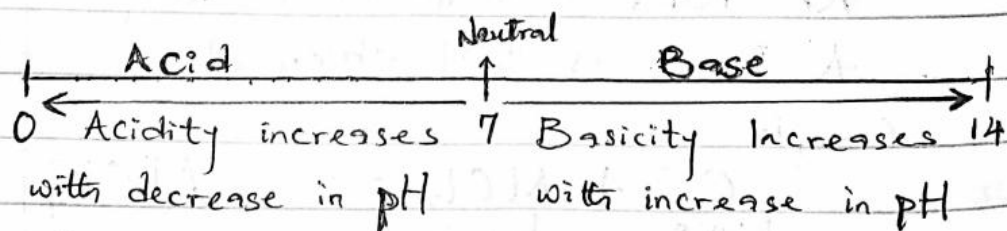
Hydrogen ion — H^+

Hydroxonium ion — H_3O^+

pH SCALE

The pH Scale gives a clear explanation of how acidic or basic a substance or solution can be. It ranges from 0 to 14.

A low pH indicates acidity, while a high pH indicates basicity or alkalinity. A pH is only neutral when it is 7.



Below 7 → Acidic

Above 7 → Basic

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Note: The pH scale does not have fixed limit, so it is possible to have substances or solutions with pH greater than 14 and less than 0.

PH MEASUREMENT/CALCULATIONS.

Whenever you see the letter "p" in front of a value, (pH or pOH), it means we are dealing with a negative logarithm of the value.

For Acidic Solutions

$$\text{pH} = -\text{Log} [\text{H}^+] \text{ or } -\text{Log} [\text{H}_3\text{O}^+]$$

But to get the H^+ , we take the inverse log of -pH

$$[\text{H}^+] = 10^{-\text{pH}}$$

For Basic Solutions

$$\text{pOH} = -\text{Log} [\text{OH}^-]$$

But to get the OH^- , we take the inverse log of -pOH.

$$[\text{OH}^-] = 10^{-\text{pOH}}$$

$$[\text{H}^+][\text{OH}^-] = 1 \times 10^{-14} = K_w \rightarrow \text{Auto Ionization Constant}$$

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For any aqueous solution at 25°C
 $\text{pH} + \text{pOH} = 14$

Solved Examples

1. What is the pH of a conc. $\text{Ca}(\text{OH})_2$ solution with concentration of 0.025M ?

Solution

$$\text{pH} + \text{pOH} = 14$$

$$\text{pH} = 14 - \text{pOH}$$

$$\text{But } \text{pOH} = -\text{Log}[0.025]$$

$$\text{pOH} = 1.6$$

$$\therefore \text{pH} = 14 - 1.6$$

$$\text{pH} = 12.4$$

$\therefore$ It is highly Basic.

NOTE: We used pOH because the solution is basic from the question. We only solved to know the extent of its basicity.

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2. What is the pH of a solution if the H_3O^+ is $2.5 \times 10^{-4} M$?

Solution

$$pH = -\log [2.5 \times 10^{-4}]$$
$$= 3.6$$

3. If the pOH of a solution is 3.2. Calculate its pH.

Solution

$$pH + pOH = 14$$

$$pH = 14 - pOH$$

$$= 14 - 3.2$$

$$= 10.8$$

4. If the pOH is 3.8, what is the $[OH^-]$ conc?

Solution

$$[OH^-] = 10^{-pOH}$$

$$= 10^{-3.8}$$

$$= 1.58 \times 10^{-4} M$$

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